Light

GCSE Computer Science — Computational Thinking (Set 2)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A student needs to create a recipe app. They identify these sub-tasks: store recipes in a database, allow users to search by ingredient, display cooking instructions, set timers for cooking steps.
What computational thinking technique has the student used?

- ○ **A.** Decomposition — breaking a complex problem into smaller, manageable sub-problems
- ○ **B.** Compilation — converting code to machine language
- ○ **C.** Debugging — finding and fixing errors
- ○ **D.** Formatting — arranging text on screen

Q2 (1 mark)

A weather app shows temperature, rainfall chance, and wind speed. It does not show barometric pressure, dew point, or UV index readings from the weather station.
What computational thinking technique does this demonstrate?

- ○ **A.** Data backup — copying files for safety
- ○ **B.** Abstraction — hiding unnecessary details to focus on essential information
- ○ **C.** Data recovery — restoring lost files
- ○ **D.** Data corruption — damaging stored information

Q3 (1 mark)

To make a cup of tea, you follow these steps: boil water, put teabag in cup, pour water over teabag, wait 3 minutes, remove teabag, add milk if desired.
What does this sequence of steps represent?

- ○ **A.** A variable
- ○ **B.** A constant
- ○ **C.** An algorithm
- ○ **D.** A data type

Q4 (1 mark)

A programmer realises that calculating a student's average mark, a cricketer's batting average, and a company's average sale value all use the same formula: sum the values and divide by the count.
What computational thinking technique identified this?

- ○ **A.** Encryption

- ○ **B.** Compression
- ○ **C.** Defragmentation
- ○ **D.** Pattern recognition

Q5 (1 mark)

A teacher has an unsorted pile of 30 exam papers. To find a specific student's paper, they check each paper from the top one by one.
What search algorithm is the teacher using?

- ○ **A.** Linear search — checking each item sequentially until the target is found
- ○ **B.** Binary search — repeatedly halving the search space
- ○ **C.** Bubble sort — swapping adjacent items
- ○ **D.** Merge sort — dividing and merging

Q6 (1 mark)

A phone book lists names alphabetically. To find 'Smith', you open to the middle, see 'M', know 'S' comes after, then search the second half using the same technique.
What search algorithm is this?

- ○ **A.** Insertion sort
- ○ **B.** Binary search — repeatedly dividing the search space in half based on comparison with the middle element
- ○ **C.** Bubble sort

- ○ **D.** Linear search

Q7 (1 mark)

Students line up by height. Starting from the left, each pair compares heights and swaps if the taller person is on the left. This repeats until no swaps are needed.
What sorting algorithm does this describe?

- ○ **A.** Merge sort
- ○ **B.** Binary search
- ○ **C.** Bubble sort — repeatedly comparing adjacent pairs and swapping if out of order
- ○ **D.** Linear search

Q8 (1 mark)

When playing cards, a player picks up one card at a time and slots it into the correct position in their hand, keeping the hand sorted as they go.
What sorting algorithm does this resemble?

- ○ **A.** Quick sort
- ○ **B.** Merge sort
- ○ **C.** Bubble sort
- ○ **D.** Insertion sort — building a sorted sequence one element at a time by inserting each new element in its correct position

Q9 (1 mark)

To sort a deck of cards, you split it in half repeatedly until you have individual cards, then combine pairs in order, then combine those pairs, until the whole deck is sorted.
What sorting algorithm is this?

- ○ **A.** Merge sort — dividing into smaller pieces then merging them back in sorted order
- ○ **B.** Linear search
- ○ **C.** Binary search
- ○ **D.** Bubble sort

Q10 (1 mark)

A programmer creates a table with columns for each variable and rows for each line of code execution, filling in how values change.
What is this technique called?

- ○ **A.** A truth table
- ○ **B.** A trace table — tracking variable values through program execution
- ○ **C.** A hash table
- ○ **D.** A routing table

Q11 (1 mark)

In a flowchart, an oval shape appears at the very beginning and very end of the diagram.
What do these ovals represent?

- ○ **A.** Decisions where the flow branches

- ○ **B.** Input or output operations
- ○ **C.** Start and end points (terminators) of the algorithm
- ○ **D.** Processing or calculation steps

Q12 (1 mark)

A programmer writes:

```
IF temperature > 30 THEN
    OUTPUT "Hot day"
ELSE
    OUTPUT "Pleasant day"
ENDIF
```

What type of notation is this?

- ○ **A.** Assembly language
- ○ **B.** Machine code
- ○ **C.** HTML markup
- ○ **D.** Pseudocode — structured English describing algorithm logic

Q13 (2 marks)

A list contains 500 unsorted items. On average, how many comparisons would linear search need to find an item that exists in the list?

- ○ **A.** 250 comparisons — on average, the item will be found halfway through the list
- ○ **B.** 500 comparisons — every item must always be checked

- ○ **C.** 1 comparison — the item might be first
- ○ **D.** 9 comparisons — using $\log_2(500)$

Q14 (2 marks)

A programmer tries to use binary search on a list but gets incorrect results.
What is the most likely cause?

- ○ **A.** The list is too short for binary search
- ○ **B.** The list is not sorted — binary search requires sorted data to work correctly because it eliminates half the search space based on comparisons with the middle element
- ○ **C.** The list contains duplicate values
- ○ **D.** The list uses integers instead of strings

Q15 (2 marks)

Bubble sort is applied to [7, 3, 5, 2]. After the first complete pass, what is the list?

- ○ **A.** [2, 3, 5, 7] — completely sorted
- ○ **B.** [7, 3, 5, 2] — unchanged
- ○ **C.** [2, 7, 5, 3] — reversed
- ○ **D.** [3, 5, 2, 7] — only the largest element is guaranteed to be in its final position

Q16 (2 marks)

Consider this pseudocode:

```
total ← 0
FOR i ← 1 TO 3
    total ← total + i
ENDFOR
```

What is the final value of total?

- ○ **A.** 0 — the loop does not change total
- ○ **B.** 3 — only the last value is stored
- ○ **C.** 6 — the sum of 1 + 2 + 3
- ○ **D.** 9 — each value is doubled

Q17 (2 marks)

A flowchart has a decision diamond asking 'Is password correct?' with 'Yes' leading to 'Grant access' and 'No' leading back to 'Enter password'. What programming construct does this represent?

- ○ **A.** A sequence — steps in order
- ○ **B.** A function call
- ○ **C.** A variable declaration
- ○ **D.** A loop — the 'No' path returns to an earlier step, creating repetition until the condition is met

Q18 (2 marks)

Merge sort is sorting [4, 1, 3, 2]. After the divide phase completes, what does the data look like?

- ○ **A.** [1, 2, 3, 4] — already sorted
- ○ **B.** [4], [1], [3], [2] — four separate single-element lists
- ○ **C.** [4, 1, 3, 2] — unchanged as one list
- ○ **D.** [2, 3, 1, 4] — randomly rearranged

Q19 (2 marks)

A car's dashboard shows speed, fuel level, and engine temperature. It does not show the hundreds of sensor readings the car's computer processes internally.
Why is this abstraction beneficial?

- ○ **A.** It saves money on display screens
- ○ **B.** It prevents the car from functioning properly
- ○ **C.** It hides complex internal data, presenting only information the driver needs to make decisions; this reduces cognitive load and improves usability
- ○ **D.** It makes the car slower

Q20 (2 marks)

A team of four programmers must build an e-commerce website in 8 weeks. They decompose it into: product catalogue, shopping basket, payment processing, and user accounts.
How does decomposition help them meet the deadline?

- ○ **A.** All four must work on each component together

- ○ **B.** Decomposition slows down development
- ○ **C.** Each programmer can work on a different component simultaneously, reducing the elapsed time compared to sequential development
- ○ **D.** Only one component can be built at a time

Q21 (2 marks)

Insertion sort is applied to [5, 1, 4, 2]. After the second element (1) has been inserted into its correct position, what does the list look like?

- ○ **A.** [1, 5, 4, 2] — the first two elements are now sorted relative to each other
- ○ **B.** [5, 1, 4, 2] — unchanged
- ○ **C.** [1, 2, 4, 5] — completely sorted
- ○ **D.** [2, 4, 1, 5] — randomly rearranged

Q22 (2 marks)

A programmer identifies that user login, admin login, and API authentication all require: accepting credentials, checking against stored data, and returning success or failure.
What benefit does recognising this pattern provide?

- ○ **A.** Each system must still be completely separate
- ○ **B.** A single authentication module can be designed that handles all three cases, reducing code duplication and ensuring consistent security
- ○ **C.** The pattern recognition is interesting but has no practical benefit

○ **D.** It means none of the systems need authentication

Q23 (2 marks)

A list needs to be searched once. The list has 20 items and is unsorted. Is it worth sorting the list to use binary search instead of linear search?

○ **A.** Yes — binary search is always faster

○ **B.** Yes — sorting is always instant

○ **C.** No — for a single search on a small list, sorting (which takes many comparisons) plus binary search takes more work than just doing linear search directly

○ **D.** No — binary search does not work on small lists

Q24 (3 marks)

A programmer wrote code to count how many times a value appears in a list, but it returns incorrect results:

```
count ← 1
FOR i ← 0 TO LENGTH(list) - 1
    IF list[i] = target THEN
        count ← count + 1
    ENDIF
ENDFOR
RETURN count
```

Using a trace table, identify the bug.

○ **A.** The count should start at 0, not 1 — starting at 1 means the result is always one higher than the actual count because an initial count of 1 is added before any matches are found

○ **B.** The loop should start at 1

- ○ **C.** The comparison operator is wrong
- ○ **D.** The loop runs backwards

Q25 (3 marks)

A programmer must sort 50,000 items. They are deciding between insertion sort and merge sort.
Which should they choose and why?

- ○ **A.** Insertion sort — it is simpler to implement
- ○ **B.** Insertion sort — it always uses less memory
- ○ **C.** Either algorithm performs identically on all data sizes
- ○ **D.** Merge sort — insertion sort's $O(n^2)$ means roughly 2.5 billion comparisons while merge sort's $O(n \log n)$ means roughly 780,000; the efficiency difference is crucial for large datasets

Q26 (3 marks)

A sorted database contains 1 billion records (1,000,000,000). How many comparisons would binary search need at most to find any record?

- ○ **A.** 1,000,000,000 comparisons — every record must be checked
- ○ **B.** 500,000,000 comparisons — half the records on average
- ○ **C.** 1,000 comparisons — the cube root of the total
- ○ **D.** About 30 comparisons — $\log_2(1 \text{ billion}) \approx 30$; each comparison halves the search space, so even enormous datasets need only logarithmic comparisons

Q27 (3 marks)

A developer is documenting an algorithm that has complex mathematical calculations but simple control flow (just a sequence of formulas).
Should they use a flowchart or pseudocode?

- ○ **A.** Flowchart — it is always superior for any algorithm
- ○ **B.** Neither representation can handle mathematical calculations
- ○ **C.** Pseudocode — it expresses mathematical operations naturally and translates directly to code; flowcharts excel at control flow but become cluttered with extensive calculations in rectangles
- ○ **D.** Flowchart — mathematical formulas cannot appear in pseudocode

Q28 (3 marks)

A programmer needs to find the second largest number in a list without sorting it.
Which approach demonstrates good algorithmic thinking?

- ○ **A.** Track two variables: 'largest' and 'secondLargest'; scan through the list once, updating both appropriately when larger values are found — this solves the problem in a single pass with O(n) efficiency
- ○ **B.** Sort the entire list and take the second element
- ○ **C.** Check every possible pair of numbers
- ○ **D.** Randomly select numbers until the second largest is found

Q29 (3 marks)

A hospital system needs to manage patient records, appointments, prescriptions, and billing. The team decomposes it into four modules

with defined interfaces.
Why is defining interfaces between modules important?

- ○ **A.** Interfaces make the system slower
- ○ **B.** Interfaces prevent modules from communicating
- ○ **C.** Interfaces are optional and provide no benefit
- ○ **D.** Interfaces define how modules communicate without exposing internal details; this enables independent development and testing, allows modules to be updated or replaced without affecting others, and reduces bugs from dependencies on implementation details

Q30 (3 marks)

A student claims: 'Computational thinking is only needed at the start of a project to break down the problem. After that, you just write code.' Evaluate this claim.

- ○ **A.** The claim is correct — computational thinking is only for initial planning
- ○ **B.** The claim is incorrect — computational thinking applies throughout development: new sub-problems require further decomposition, implementations require algorithmic thinking, testing reveals patterns in bugs, and maintenance requires abstracting changes to minimise impact
- ○ **C.** The claim is correct — coding does not require any thinking
- ○ **D.** The claim is partially correct — only abstraction continues beyond initial planning

Total: **55** marks • Time: **30 min**

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